

Impact of Occlusal Schemes on Implant-Supported Overdentures

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ABSTRACT

Background: The occlusal scheme is a crucial factor in the success and patient satisfaction of implant-supported overdentures. Different occlusal schemes, such as balanced, lingualized, and monoplane occlusion, can influence the functional and biomechanical outcomes. **Materials and Methods:** A randomized controlled trial was conducted with 60 edentulous patients who received mandibular implant-supported overdentures. The patients were divided into three groups based on the occlusal scheme: Group A (balanced occlusion), Group B (lingualized occlusion), and Group C (monoplane occlusion), with 20 patients in each group. All patients received two implants in the mandibular canine regions. The overdentures were fabricated and adjusted according to the assigned occlusal scheme. Clinical assessments, including bite force measurement and masticatory efficiency, were performed at baseline, 3 months, and 6 months post-insertion. Patient satisfaction was evaluated using a validated questionnaire. **Results:** At 6 months, Group A (balanced occlusion) exhibited the highest mean bite force ($200 \text{ N} \pm 20$), followed by Group B (lingualized occlusion) with $180 \text{ N} \pm 15$, and Group C (monoplane occlusion) with $160 \text{ N} \pm 10$. Masticatory efficiency was significantly higher in Group A ($80\% \pm 5$) compared to Group B ($75\% \pm 4$) and Group C ($70\% \pm 3$). Patient satisfaction scores were highest in Group A (9.0 ± 0.5), followed by Group B (8.5 ± 0.4) and Group C (8.0 ± 0.3). There were statistically significant differences between the groups in terms of bite force, masticatory efficiency, and patient satisfaction ($P < 0.05$). **Conclusion:** Balanced occlusion provided superior functional outcomes and higher patient satisfaction for implant-supported overdentures compared to lingualized and monoplane occlusion schemes.

KEYWORDS: *Balanced occlusion, bite force, implant-supported overdentures, lingualized occlusion, masticatory efficiency, monoplane occlusion, occlusal schemes, patient satisfaction*

INTRODUCTION

The rehabilitation of edentulous patients with implant-supported overdentures has become a widely accepted treatment modality due to its enhanced stability and retention compared to conventional dentures. One of the critical factors influencing the success of implant-supported overdentures is the occlusal scheme, which can significantly affect the functional and biomechanical outcomes of the prosthesis.^[1] The choice of occlusal scheme plays a vital role in distributing occlusal forces, which in turn impacts the performance

and longevity of the overdentures as well as patient satisfaction.^[2]

Balanced occlusion, lingualized occlusion, and monoplane occlusion are commonly used occlusal schemes in prosthodontics. Balanced occlusion aims to achieve simultaneous bilateral contact during centric

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and eccentric movements, thereby distributing forces evenly.^[3] Lingualized occlusion, on the other hand, focuses on achieving a harmonious relationship between the maxillary palatal cusps and the mandibular fossae, providing a balance between stability and functional efficiency.^[4] Monoplane occlusion utilizes flat occlusal surfaces to minimize lateral forces and simplify denture adjustments.^[5]

Previous studies have shown varying results regarding the effectiveness of these occlusal schemes. Some research suggests that balanced occlusion provides better masticatory performance and patient satisfaction compared to monoplane and lingualized occlusion.^[6,7]

MATERIALS AND METHODS

A total of 60 edentulous patients were recruited for the study. The inclusion criteria were:

1. Completely edentulous mandible.
2. Good general health with no contraindications for implant surgery.
3. Adequate bone volume in the mandibular canine regions is to accommodate implants.
4. Willingness to participate and comply with study protocols.

Exclusion criteria include:

1. History of systemic conditions that could affect bone healing (e.g., uncontrolled diabetes, osteoporosis).
2. Previous implant treatment in the mandible.
3. Significant maxillofacial deformities.
4. Psychological or neurological disorders affect the ability to follow instructions.

Participants were randomly assigned to one of three groups (20 patients each) based on the occlusal scheme of their implant-supported overdentures:

- Group A: Balanced occlusion.
- Group B: Lingualized occlusion.
- Group C: Monoplane occlusion.

Clinical assessments were performed at baseline (pre-insertion), 3 months, and 6 months

post-insertion. The following parameters were evaluated:

- **Bite Force Measurement:** Measured using a bite force transducer (Occlusal Force Meter GM10, Nagano Keiki, Japan) at the first molar region bilaterally. The average of three measurements was recorded.
- **Masticatory Efficiency:** Assessed using a colorimetric method with a standardized test food (two-color chewing gum) to determine the degree of color mixing after a specified number of chewing strokes.

Patient satisfaction was evaluated at 6 months post-insertion using a validated questionnaire (OHIP-EDENT) consisting of 19 items covering aspects such as comfort, stability, ease of cleaning, and overall satisfaction. Responses were recorded on a Likert scale from 1 (very dissatisfied) to 10 (very satisfied).

Statistical analysis

Data were analyzed using SPSS software (version 25.0, IBM Corp., USA).

RESULTS

Bite force measurement

The mean bite force values for each group at baseline, 3 months, and 6 months are presented in Table 1. At 6 months, Group A (balanced occlusion) exhibited the highest mean bite force ($200 \text{ N} \pm 20$), followed by Group B (lingualized occlusion) with $180 \text{ N} \pm 15$, and Group C (monoplane occlusion) with $160 \text{ N} \pm 10$. The increase in bite force from baseline to 6 months was statistically significant for all groups ($P < 0.05$).

Masticatory efficiency

Masticatory efficiency, as assessed by the colorimetric method, showed significant improvements in all groups from baseline to 6 months ($P < 0.05$). Group A demonstrated the highest masticatory efficiency at 6 months ($80\% \pm 5$), compared to Group B ($75\% \pm 4$) and Group C ($70\% \pm 3$). The differences between the groups were statistically significant ($P < 0.05$) [Table 2].

Table 1: Mean Bite Force (N) at Baseline, 3 Months, and 6 Months

Time Point	Group A (Balanced Occlusion)	Group B (Lingualized Occlusion)	Group C (Monoplane Occlusion)
Baseline	100 N $\pm$ 15	95 N $\pm$ 10	90 N $\pm$ 12
3 Months	150 N $\pm$ 18	140 N $\pm$ 13	130 N $\pm$ 15
6 Months	200 N $\pm$ 20	180 N $\pm$ 15	160 N $\pm$ 10

Table 2: Masticatory Efficiency (%) at Baseline, 3 Months, and 6 Months

Time Point	Group A (Balanced Occlusion)	Group B (Lingualized Occlusion)	Group C (Monoplane Occlusion)
Baseline	50% $\pm$ 7	48% $\pm$ 6	45% $\pm$ 5
3 Months	65% $\pm$ 6	62% $\pm$ 5	58% $\pm$ 4
6 Months	80% $\pm$ 5	75% $\pm$ 4	70% $\pm$ 3

Table 3: Patient Satisfaction Scores at 6 Months

Group	Satisfaction score (Mean±SD)
Group A (Balanced Occlusion)	9.0±0.5
Group B (Lingualized Occlusion)	8.5±0.4
Group C (Monoplane Occlusion)	8.0±0.3

Patient satisfaction

Patient satisfaction scores were highest in Group A at 6 months (9.0 ± 0.5), followed by Group B (8.5 ± 0.4) and Group C (8.0 ± 0.3). The differences in patient satisfaction scores between the groups were statistically significant ($P < 0.05$) [Table 3].

DISCUSSION

The superior bite force observed in the balanced occlusion group aligns with previous studies that have highlighted the advantages of balanced occlusion in evenly distributing occlusal forces and enhancing masticatory performance.^[1,2] The mean bite force in Group A ($200 \text{ N} \pm 20$) at 6 months was significantly higher than in Group B ($180 \text{ N} \pm 15$) and Group C ($160 \text{ N} \pm 10$). This suggests that the simultaneous bilateral contact achieved in balanced occlusion may contribute to more effective force distribution, reducing the load on individual implants and improving overall bite strength.^[3]

Masticatory efficiency was also highest in the balanced occlusion group ($80\% \pm 5$) compared to the lingualized ($75\% \pm 4$) and monoplane occlusion groups ($70\% \pm 3$). These results are consistent with the findings of Rizzatti-Barbosa CM *et al.*, who reported improved masticatory performance with balanced occlusion due to its ability to maintain stable and efficient contacts during mastication.^[4] The enhanced masticatory efficiency in balanced occlusion can be attributed to the comprehensive tooth contacts that facilitate better food breakdown and chewing efficiency.^[5]

Patient satisfaction scores were significantly higher in the balanced occlusion group (9.0 ± 0.5) compared to the lingualized (8.5 ± 0.4) and monoplane occlusion groups (8.0 ± 0.3). This is in line with studies by Kapur^[6] and van Kampen FMC *et al.*,^[7] which found that patients

with balanced occlusion overdentures reported higher levels of comfort and satisfaction due to the stability and functional efficiency of their prostheses. The balanced occlusal scheme appears to provide a more natural and comfortable biting experience, contributing to greater patient acceptance and satisfaction.

CONCLUSION

This study highlights the superiority of balanced occlusion in enhancing bite force, masticatory efficiency, and patient satisfaction in implant-supported overdentures. The findings suggest that balanced occlusion should be the preferred occlusal scheme for clinicians aiming to improve the functional and subjective outcomes of implant-supported overdentures.

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Conflicts of interest

There are no conflicts of interest.

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